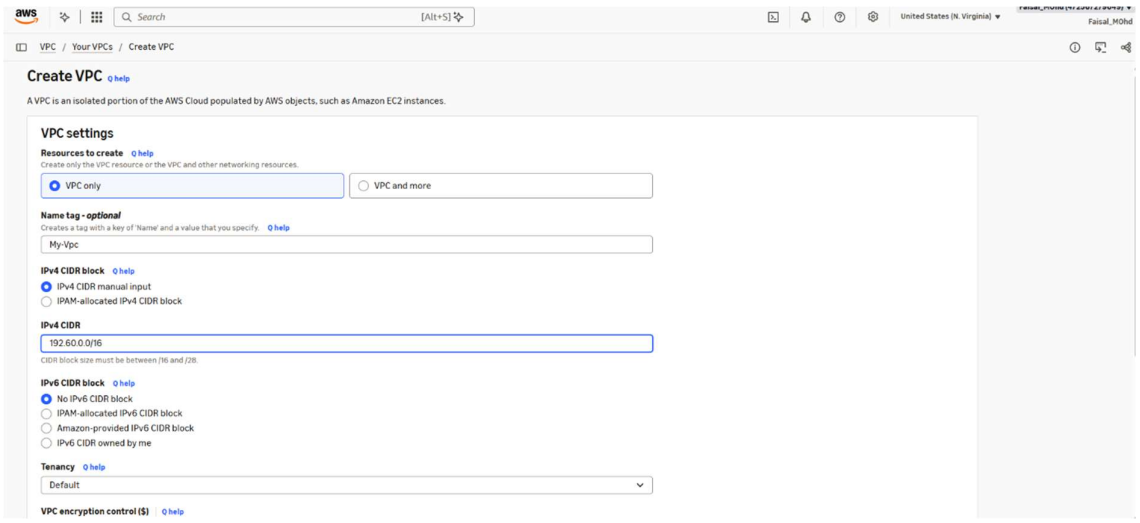
Auto Scaling Group Task

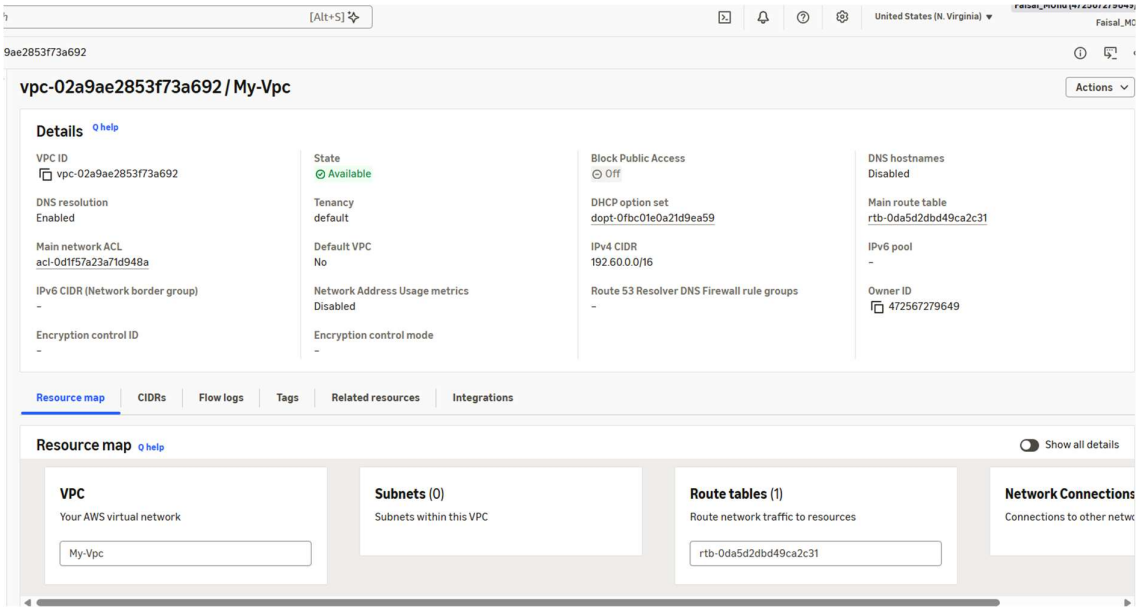
1.Create one VPC in N. Virginia region.

Step 1:- Open the VPC service section, go to region and change it to n. Vigrinia region.

Step 2:- Create a VPC and custom IPv4 CIDR. Then click create

Step 3:- Now a VPC is created in N.Virginia region.





2. Create two subnets: one public subnet and one private subnet.

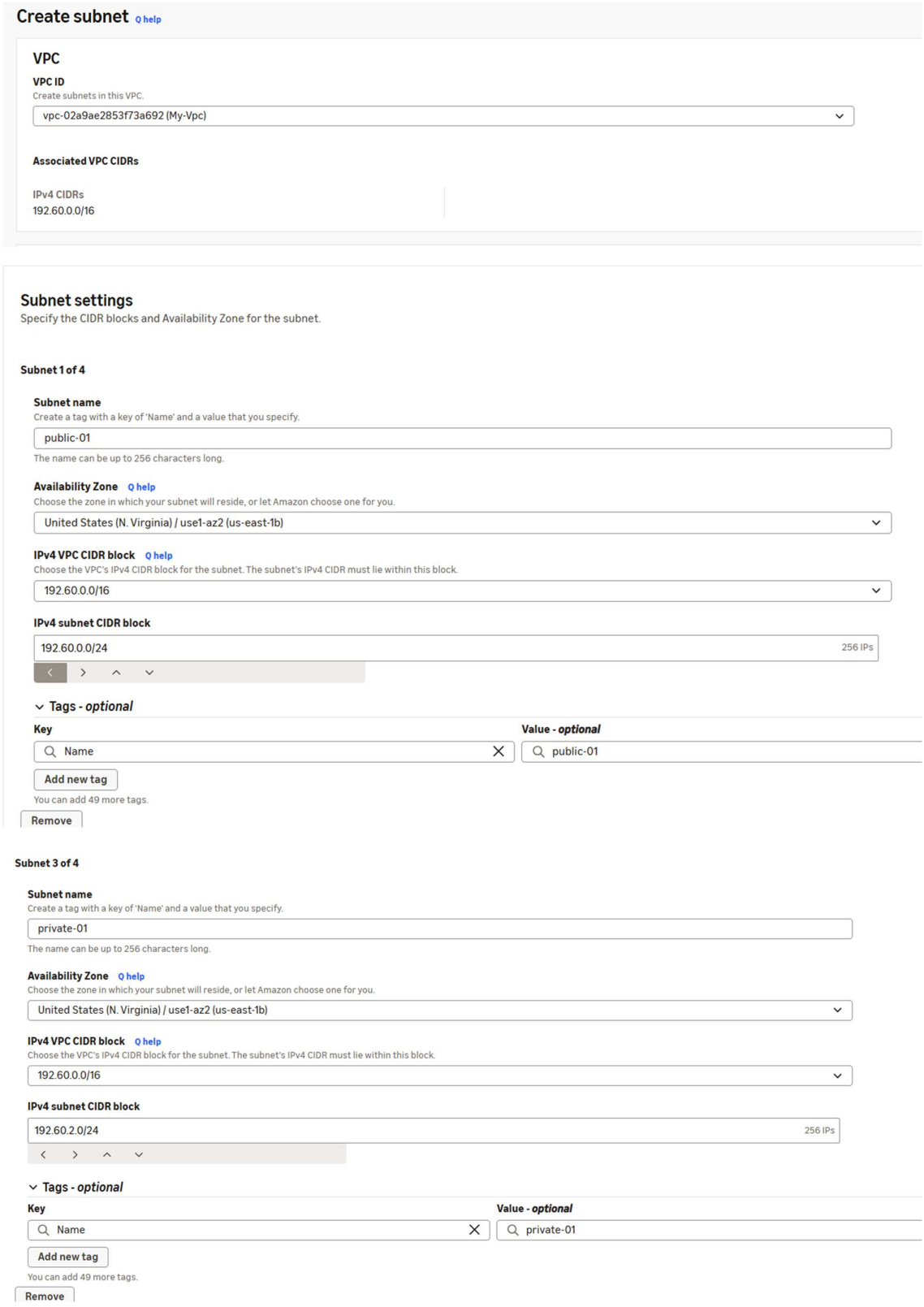
Step 1:- Go to subnets section and click on create subnet.

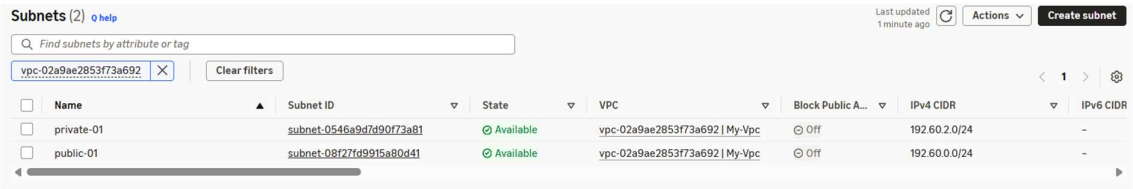
Step 2:- Now select the VPC and scroll down and name the subnets and assign the CIDR range.

Step 3:- Similarly there is option to add more subnets at a time.

Step 4:- Name the subnets and assign the subnets CIDR range and subnet masking.

Step 5:- Click create then one private and one public subnets are created.



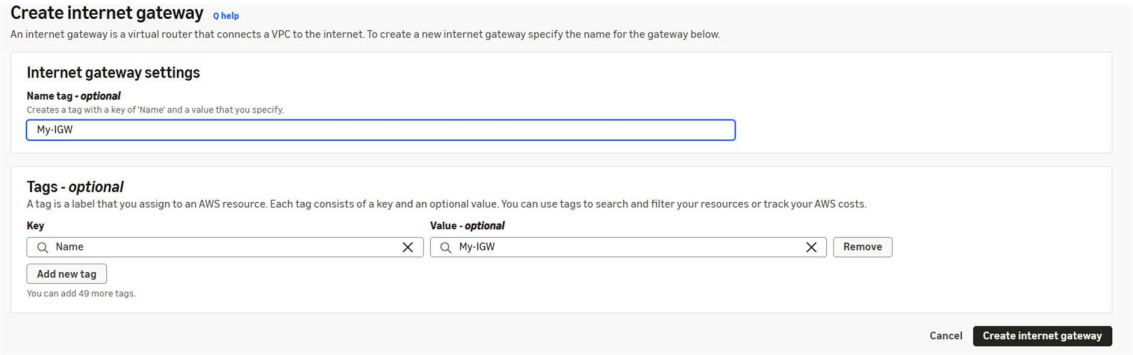


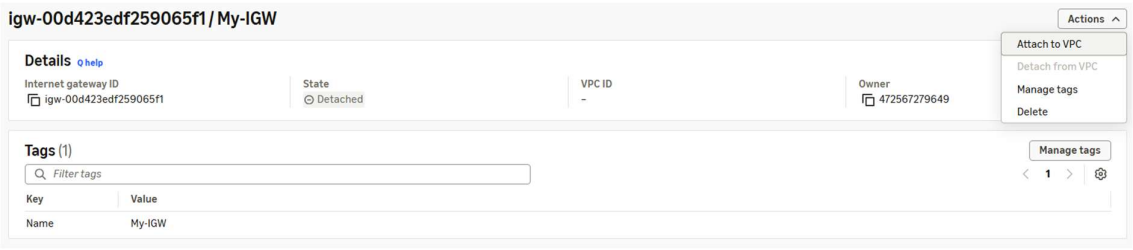
3. Attach an IGW to the VPC.

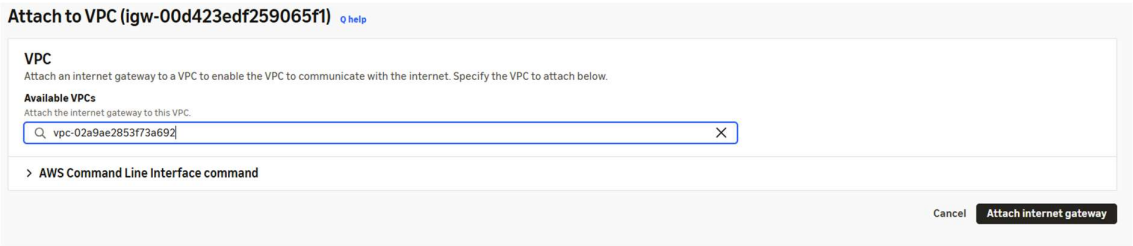
Step 1:- Go to Internet gateway section and create a new IGW.

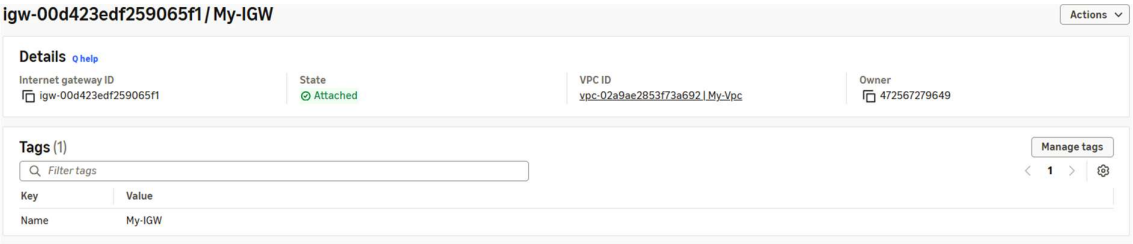
Step 2:- After creating the IGW select the IGW and go to actions. There click on Attach VPC.

Step 3:-Now select the VPC and click attach. Now the VPC is attached to IGW.







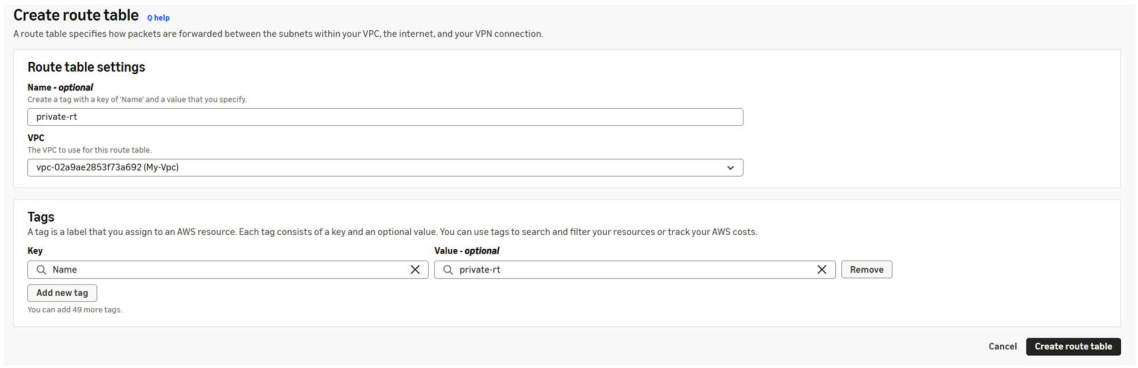


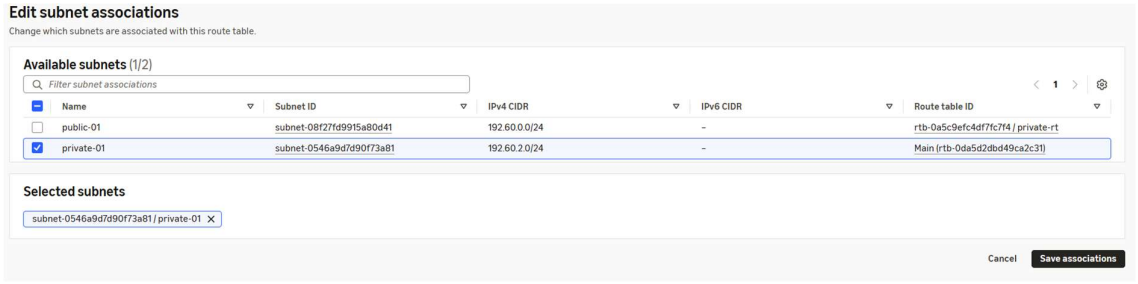
4. Create one public route table (RT) and one private route table.

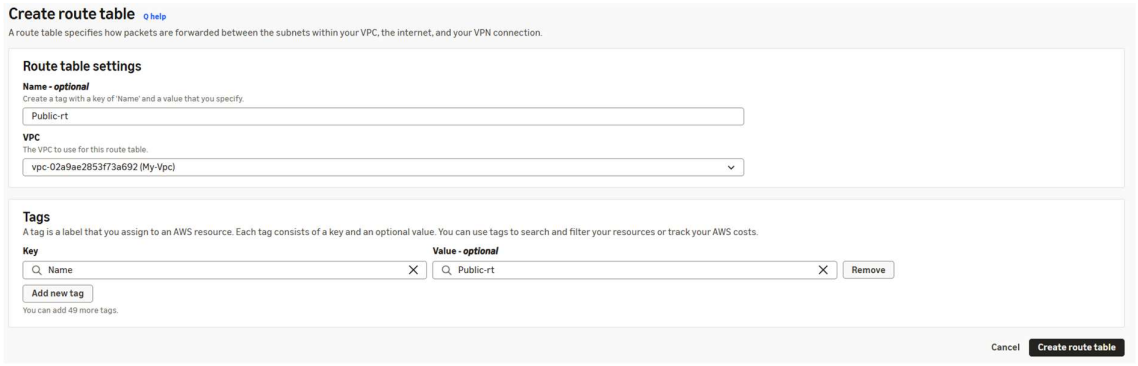
Step 1:- Go to Route tables section and create two route tables.

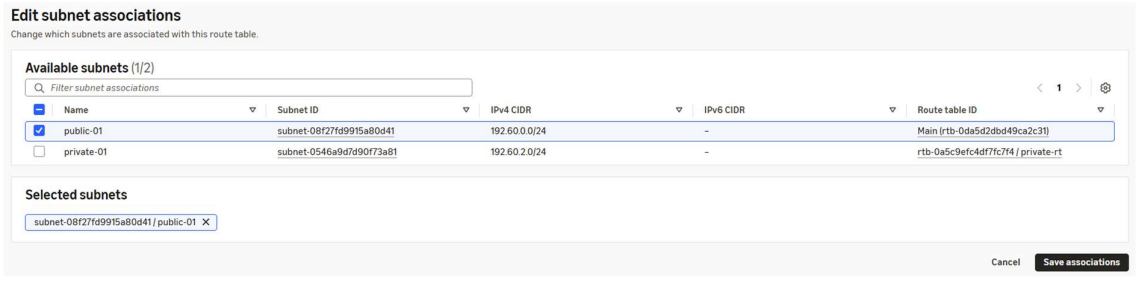
Step 2:- In public route table add public subnet in the subnet associations and IGW in the routes.

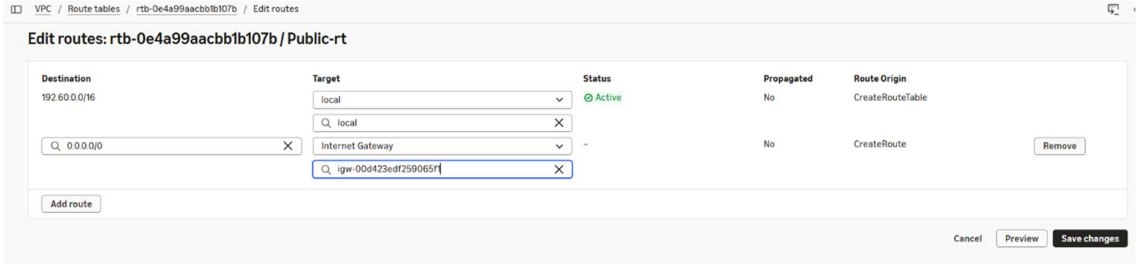
Step 3:- Create a private route table and add the private subnets in the subnets associations.









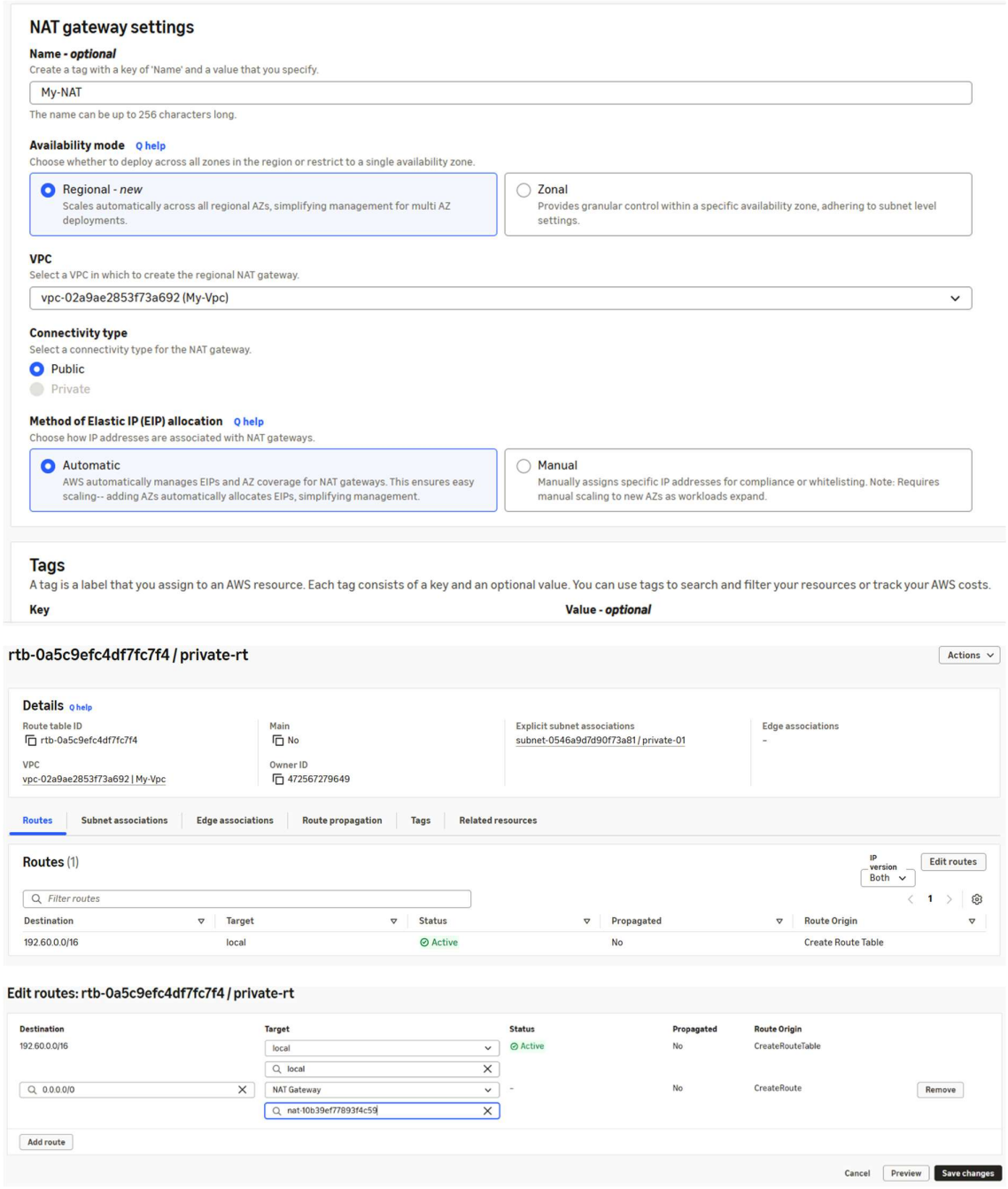


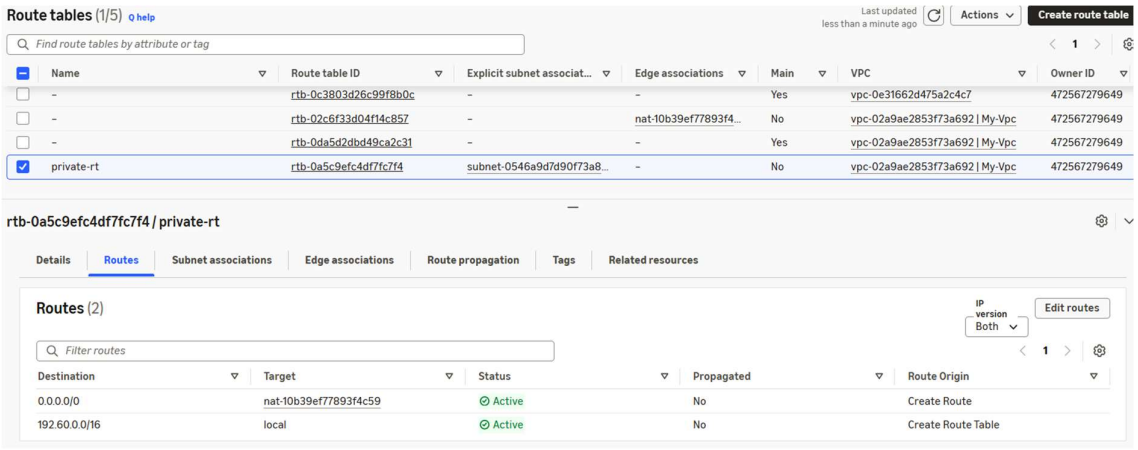
5. Deploy a NAT gateway in the public subnet and attach the NAT gateway to the private subnet.

Step 1:- Create the NAT gateways in the VPC services. Assign the VPC when the Nat gateways is created.

Step 2:- Then go to private-rt and add the NAT gateway in the routes.

Step 3:- Now the NAT gateway is assigned to the Private route table.



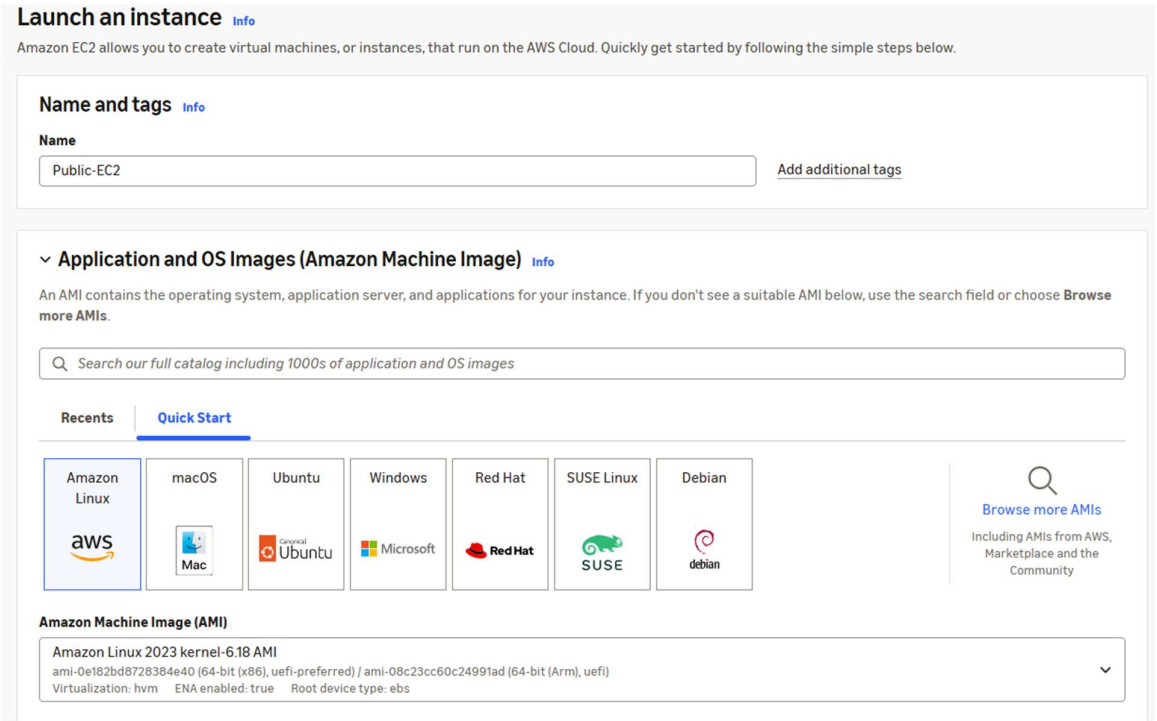


6. Create two instances, one in the public subnet and one in the private subnet.

Step 1:- First create a Private instance. Go to launch instance and change the VPC and select private subnet.

Step 2:- Now create a instance with the VPC and public subnet. And launch the instance.

Step 3:- One private and One Public instance is created.



Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

EC2 | Create new key pair

Network settings Info

VPC - *required* Info

vpc-02a9ae2853f73a692 (My-Vpc)
192.60.0.0/16

Subnet Info

subnet-08f27fd9915a80d41 public-01 | Create new subnet
VPC: vpc-02a9ae2853f73a692 Owner: 472567279649 Availability Zone: us-east-1b (use1-az2)
Zone type: Availability Zone IP addresses available: 251 CIDR: 192.60.0.0/24)

Auto-assign public IP Info

Disable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

○ Create security group ◉ Select existing security group

Common security groups Info

Select security groups | Compare security group rules

default sg-052d09390de365dc9 ×
VPC: vpc-02a9ae2853f73a692

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Instance summary for i-0afdb132f368bca8a (Public-Ec2) Info

Connect | Instance state | Actions

Updated less than a minute ago

Instance ID i-0afdb132f368bca8a	Public IPv4 address 3.94.61.78 | open address	Private IPv4 addresses 192.60.0.96
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-192-60-0-96.ec2.internal	Private IP DNS name (IPv4 only) ip-192-60-0-96.ec2.internal	Answer private resource DNS name -
Instance type t3.micro	Elastic IP addresses -	Auto-assigned IP address 3.94.61.78 [Public IP]
VPC ID vpc-02a9ae2853f73a692 (My-Vpc)	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. | Learn more	IAM role -
Subnet ID subnet-08f27fd9915a80d41 (public-01)	Auto Scaling group name -	IMDSv2 Required
Instance ARN arn:aws:ec2:us-east-1:472567279649:instance/i-0afdb132f368bca8a	Managed false	Operator -

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

Private-Ec2

Add additional tags

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

Recents | Quick Start

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux | Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI
ami-09bf9dfeeb1e9adbc (64-bit (x86), uefi-preferred) / ami-0cc337c0f1a9da5ed (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

EC2

Create new key pair

Network settings Info

VPC - *required* Info

vpc-02a9ae2853f73a692 (My-Vpc)
192.60.0.0/16

Subnet Info

subnet-0546a9d7d90f73a81 private-01
VPC: vpc-02a9ae2853f73a692 Owner: 472567279649 Availability Zone: us-east-1b (use1-az2)
Zone type: Availability Zone IP addresses available: 251 CIDR: 192.60.2.0/24)

Create new subnet

Auto-assign public IP Info

Disable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

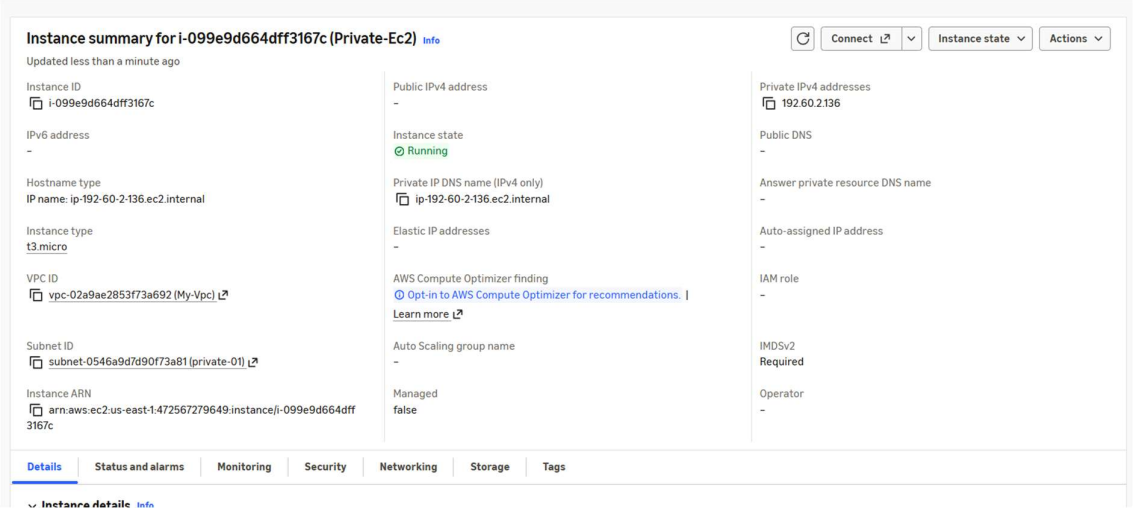
- ○ Create security group
- ◉ Select existing security group

Common security groups Info

Select security groups

Compare security group

default sg-052d09390de365dc9 ×
VPC: vpc-02a9ae2853f73a692



7. Deploy Apache server on both EC2 instances with a sample index.html file.

Step 1:- First connect with the public instance and install httpd package.

Step 2:- Start the package, then go to /var/www/html folder and create a index.html file.

Step 3:- Now if we check the port 80 of this instance we can see the custom index.html.

Step 4:- Copy the pem file from window to public instance. Change the file permissions.

Step 5:- Then open the private instance from public instance.Now install the httpd package then start the package.

Step 6:- Go to /var/www/html and create custom index.html file. To check this page we have to use curl command to fetch the response.

```
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-192-60-0-96 ~]$ sudo yum update
Amazon Linux 2023 Kernel Livepatch repository
Last metadata expiration check: 0:00:01 ago on Thu Nov 12 12:24:05 2026.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-192-60-0-96 ~]$ sudo yum install httpd -y
Last metadata expiration check: 0:00:16 ago on Thu Nov 12 12:24:05 2026.
Dependencies resolved.
================================================================================
 Package                    Architecture                Version
================================================================================
Installing:
 httpd                      x86_64                      2.4.69-1.amzn2023.0.2
Installing dependencies:
 apr                        x86_64                      1.7.5-1.amzn2023.0.4
 apr-util                   x86_64                      1.6.5-1.amzn2023.0.1
 apr-util-lmdb              x86_64                      1.6.5-1.amzn2023.0.1
 generic-logos-httpd        noarch                      18.0.0-12.amzn2023.0.3
 httpd-core                 x86_64                      2.4.69-1.amzn2023.0.2
 httpd-filesystem           noarch                      2.4.69-1.amzn2023.0.2
 httpd-tools                x86_64                      2.4.69-1.amzn2023.0.2
 libbrotli                  x86_64                      1.1.0-1.amzn2023.0.2
 mailcap                    noarch                      2.1.49-3.amzn2023.0.3
Installing weak dependencies:
 apr-util-openssl           x86_64                      1.6.5-1.amzn2023.0.1
 mod_http2                  x86_64                      1:2.4.69-1.amzn2023.0.2
```

```
[ec2-user@ip-192-60-0-96 ~]$ sudo systemctl start httpd
[ec2-user@ip-192-60-0-96 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: active (running) since Thu 2026-11-12 12:25:01 UTC; 8s ago
       Docs: man:httpd.service(8)
   Main PID: 26524 (httpd)
     Status: "Started, listening on: port 80"
      Tasks: 177 (limit: 1059)
     Memory: 13.8M
        CPU: 64ms
     CGroup: /system.slice/httpd.service
             ├─26524 /usr/sbin/httpd -DFOREGROUND
             ├─26525 /usr/sbin/httpd -DFOREGROUND
             ├─26526 /usr/sbin/httpd -DFOREGROUND
             ├─26527 /usr/sbin/httpd -DFOREGROUND
             └─26533 /usr/sbin/httpd -DFOREGROUND

Nov 12 12:25:01 ip-192-60-0-96.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server
Nov 12 12:25:01 ip-192-60-0-96.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Nov 12 12:25:01 ip-192-60-0-96.ec2.internal httpd[26524]: Server configured, listening on: port 80
[ec2-user@ip-192-60-0-96 ~]$ ^C
```

```
[ec2-user@ip-192-60-0-96 /]$ cd /var/www/html/
[ec2-user@ip-192-60-0-96 html]$ touch index.html
touch: cannot touch 'index.html': Permission denied
[ec2-user@ip-192-60-0-96 html]$ sudo touch index.html
[ec2-user@ip-192-60-0-96 html]$ chmod 777 index.html
chmod: changing permissions of 'index.html': Operation not permitted
[ec2-user@ip-192-60-0-96 html]$ sudo chmod 777 index.html
[ec2-user@ip-192-60-0-96 html]$ vi index.html
```

```
[ec2-user@ip-192-60-0-96 html]$ cat index.html

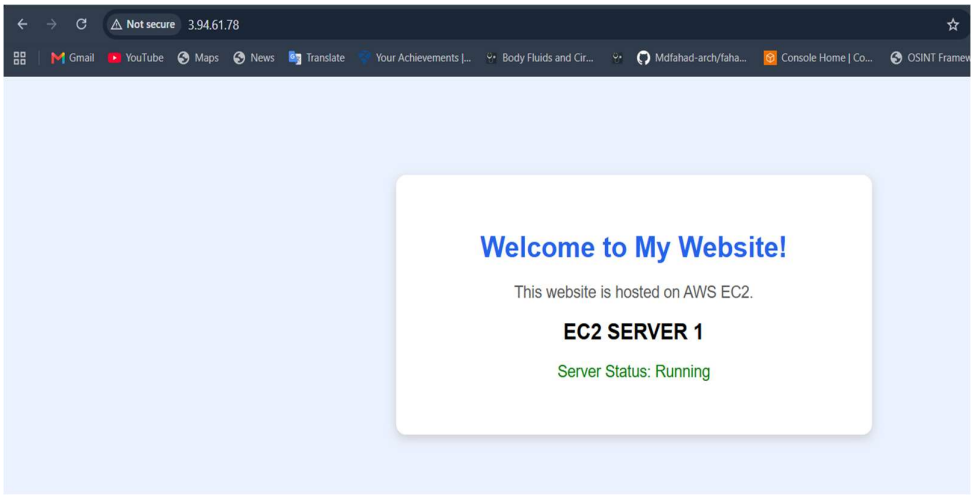
<!DOCTYPE html>
<html>
<head>
    <title>My AWS Website</title>

    <style>
        body {
            font-family: Arial;
            background: #eaf2ff;
            text-align: center;
            padding: 100px 20px;
        }

        .card {
            background: white;
            padding: 40px;
            border-radius: 12px;
            max-width: 500px;
            margin: auto;
            box-shadow: 0 4px 12px #ccc;
        }

        h1 {
            color: #2563eb;
        }

        p {
```



```
[ec2-user@ip-192-60-0-96 ~]$ touch EC2.pem
[ec2-user@ip-192-60-0-96 ~]$ vi EC2.pem
[ec2-user@ip-192-60-0-96 ~]$ chmod 400 EC2.pem
[ec2-user@ip-192-60-0-96 ~]$ ssh -i EC2.pem ec2-user@192.60.2.136
The authenticity of host '192.60.2.136 (192.60.2.136)' can't be established.
ED25519 key fingerprint is SHA256:tde6Y0D2vd1Dr1nOjHGYX49Q3D50v7CBi3F40cR8zBI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.60.2.136' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-192-60-2-136 ~]$ sudo -i
[root@ip-192-60-2-136 ~]# yum install httpd -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
================================================================================
 Package                                 Architecture                 Version
================================================================================
Installing:
 httpd                                   x86_64                       2.4.69-1.amzn2023.0.2
Installing dependencies:
 apr                                     x86_64                       1.7.5-1.amzn2023.0.4
 apr-util                                x86_64                       1.6.5-1.amzn2023.0.1
 apr-util-lmdb                           x86_64                       1.6.5-1.amzn2023.0.1
 generic-logos-httpd                     noarch                       18.0.0-12.amzn2023.0.3
 httpd-core                              x86_64                       2.4.69-1.amzn2023.0.2
 httpd-filesystem                        noarch                       2.4.69-1.amzn2023.0.2
 httpd-tools                             x86_64                       2.4.69-1.amzn2023.0.2
```

```
[root@ip-192-60-2-136 ~]# systemctl start httpd
[root@ip-192-60-2-136 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: active (running) since Thu 2026-11-12 12:36:26 UTC; 9s ago
       Docs: man:httpd.service(8)
   Main PID: 26690 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1059)
     Memory: 14.0M
        CPU: 70ms
     CGroup: /system.slice/httpd.service
             ├─26690 /usr/sbin/httpd -DFOREGROUND
             ├─26691 /usr/sbin/httpd -DFOREGROUND
             ├─26692 /usr/sbin/httpd -DFOREGROUND
             ├─26693 /usr/sbin/httpd -DFOREGROUND
             └─26694 /usr/sbin/httpd -DFOREGROUND

Nov 12 12:36:26 ip-192-60-2-136.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Nov 12 12:36:26 ip-192-60-2-136.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Nov 12 12:36:26 ip-192-60-2-136.ec2.internal httpd[26690]: Server configured, listening on: port 80
[root@ip-192-60-2-136 ~]# ^C
```

```
[root@ip-192-60-2-136 html]# touch index.html
[root@ip-192-60-2-136 html]# chmod 777 index.html
[root@ip-192-60-2-136 html]# vi index.html
[root@ip-192-60-2-136 html]# cat index.html

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Website</title>
</head>
<body>

    <h1>Welcome to My Website</h1>
    <p>This website is hosted on AWS EC2.</p>
    <h2>EC2 Server 1</h2>

</body>
</html>

[root@ip-192-60-2-136 html]#
```

```
[root@ip-192-60-2-136 html]# curl http://localhost:80

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Website</title>
</head>
<body>

    <h1>Welcome to My Website</h1>
    <p>This website is hosted on AWS EC2.</p>
    <h2>EC2 Server 1</h2>

</body>
</html>

[root@ip-192-60-2-136 html]#
```

8. Create one application load balancer and attach it to both EC2 instances.

Step 1:- Create a new public subnet in same region but different availability zone. Then create SG for HTTP port 80.

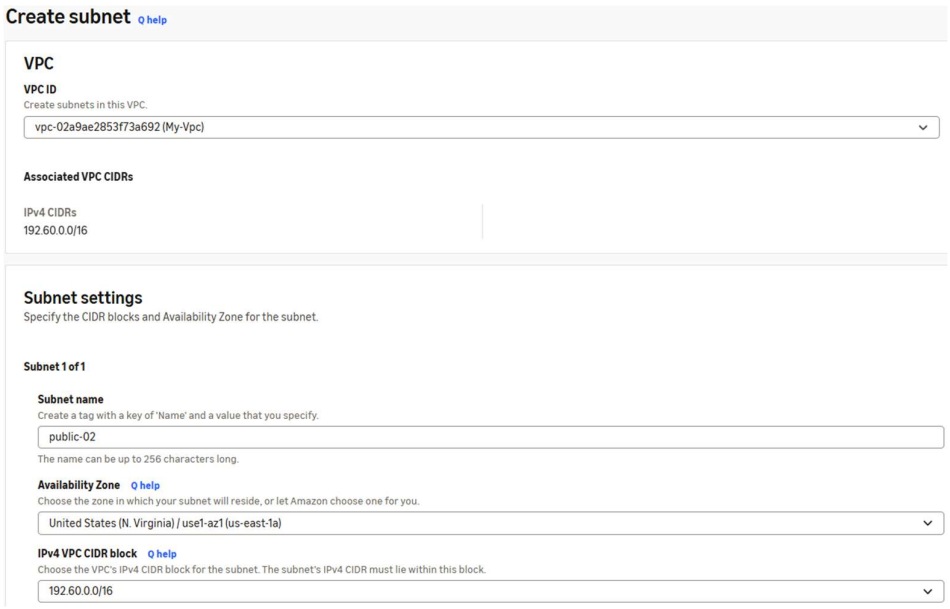
Step 2:- Create a target group select the vpc and click next there we should select the instances and click on includes as pending below.

Step 3:- Now create a Application load balancer. Select the vpc, then all the availability zone shown in the list. And change the security group, then click create.

Step 4:- After the ALB is created the insatnces move to healthy state then the ALB is successfully applied.

Step 5:- Now go to loadbalancer and copy the DNS link to open the server.

Refresh to see all the index html pages.



Create security group Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name Info

ELB-sg

Name cannot be edited after creation.

Description Info

Security group for load balancer

VPC Info

vpc-02a9ae2853f73a692 (My-Vpc)

Inbound rules Info

Type Info	Protocol Info	Port range Info	Source type Info	Source Info	Description - optional Info	
HTTP	TCP	80	Anywhere-IPv4	0.0.0.0/0		Delete

Add rule

⚠ Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - *immutable*

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

- (●) **Instances**
  Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
  Suitable for: ALB NLB GWLB
- ( ) **IP addresses**
  Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
  Suitable for: ALB NLB GWLB
- ( ) **Lambda function**
  Supports load balancing to a single Lambda function. ALB required as traffic source.
  Suitable for: ALB
- ( ) **Application Load Balancer**
  Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
  Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.

Loadbalancer

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 12/32

Protocol

Protocol for communication between the load balancer and targets.

HTTP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

- (●) **IPv4**
  Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.
- ( ) **IPv6**
  Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). Learn more

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-02a9ae2853f73a692 (My-Vpc)
192.60.0.0/16

vpc-02a9ae2853f73a692 (My-Vpc)

Create VPC

Protocol version

- (●) **HTTP1**
  Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.
- ( ) **HTTP2**
  Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.
- ( ) **gRPC**
  Send requests to targets using gRPC. Supported when the request protocol is gRPC.

Register targets - *recommended*

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2/2)

Filter instances

Instance ID	Name	State	Security groups	Zone	Private IPv4
i-0afdb132f368bca8a	Public-Ec2	Running	default	us-east-1b	192.60.0.96
i-099e9d664dff3167c	Private-Ec2	Running	default	us-east-1b	192.60.2.136

2 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

Review targets

Targets (2)

Remove all pending

Filter targets | Show only pending | 1

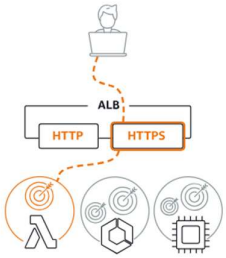
Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-0afdb132f368bca8a	Public-Ec2	80	Running	default	us-east-1b	192.60.0.96	subnet-08f27fd9915a80d41	November 12, 2026, 17:45 (U
i-099e9d664dff3167c	Private-Ec2	80	Running	default	us-east-1b	192.60.2.136	subnet-0546a9d7d90f73a81	November 12, 2026, 17:43 (U

2 pending

Cancel | Previous | Next

Load balancer types

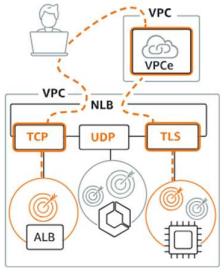
Application Load Balancer Info



Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

Create

Network Load Balancer Info



Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

Create

Gateway Load Balancer Info



Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

Create

› Classic Load Balancer - *previous generation*

Create Application Load Balancer Info

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.

› How Application Load Balancers work

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

ALB

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 3/32

Scheme | Info

Scheme can't be changed after the load balancer is created.

- (•) Internet-facing
  - Serves internet-facing traffic.
  - Has public IP addresses.
  - DNS name resolves to public IPs.
  - Requires a public subnet.
- ( ) Internal
  - Serves internal traffic.
  - Has private IP addresses.
  - DNS name resolves to private IPs.
  - Compatible with the **IPv4** and **Dualstack** IP address types.

Load balancer IP address type | Info

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

- (•) IPv4
  Includes only IPv4 addresses.
- ( ) Dualstack
  Includes IPv4 and IPv6 addresses.
- ( ) Dualstack without public IPv4
  Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with **internet-facing** load balancers only.

VPC | Info

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view target groups.

vpc-02a9ae2853f73a692 (My-Vpc)
192.60.0.0/16

IP pools | Info

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view **Pools** in the Amazon VPC IP Address Manager console.

☐ Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets | Info

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets

☑ **us-east-1a (use1-az1)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inb allow traffic.

subnet-03e1ee9a5e27f42cb (public-02)
192.60.1.0/24
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (missing)

⚠ Reachability may be impacted

☑ **us-east-1b (use1-az2)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inb allow traffic.

subnet-08f27fd9915a80d41 (public-01)
192.60.0.0/24
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (√)

Security groups Info

A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must i load balancer.

Security groups

Select up to 16 security groups

ELB-sg (sg-07a8b5c3c269b9846) ×
Inbound rule (1) Outbound rule (1)

Default action | Info

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

◉ Forward to target groups ○ Redirect to URL ○ Return fixed response

Forward to target group | Info

Choose a target group and specify routing weight or create target group.

Target group	Weight	Percent
Loadbalancer — HTTP Target type: Instance, IPv4 | Target stickiness: Off	1 0-999	100%

+ Add target group

You can add up to 4 more target groups.

Target group stickiness | Info

Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

Listener tags - *optional*

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Summary

Review and confirm your configurations. Estimate cost

Basic configuration Edit
Name: ALB
Scheme: Internet-facing
IP address type: IPv4

Network mapping Edit ⚠
VPC: vpc-02a9ae2853f73a692
Public IPv4 IPAM pool: -
Availability Zones and subnets:
- us-east-1a
  subnet-03e1ee9a5e27f42cb
  public-02
- us-east-1b
  subnet-08f27fd9915a80d41
  public-01

Security groups Edit
ELB-sg
sg-07a8b5c3c269b9846

Listeners and routing Edit
HTTP:80 | Forward to 1 target group

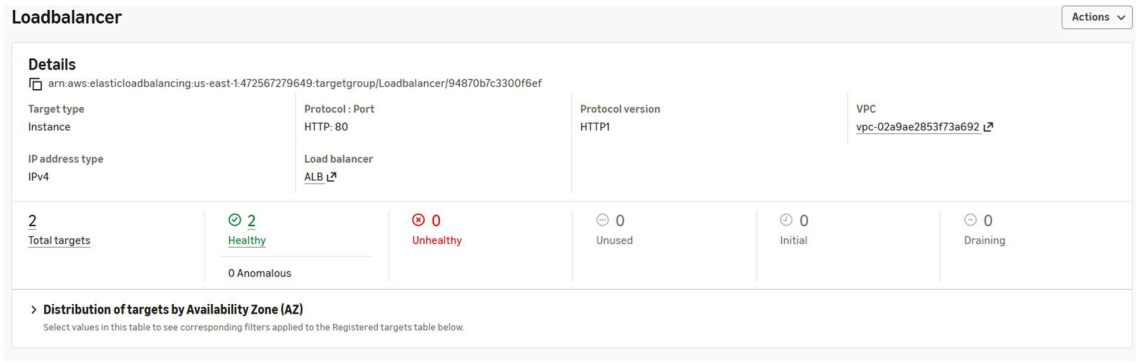
⚠ **Load balancer reachability may be impacted**
› 1 network configuration issue

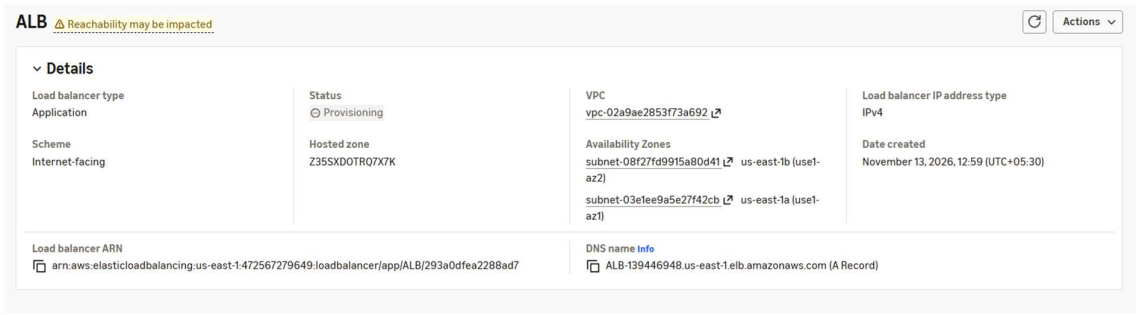
Service integrations Edit
Amazon CloudFront + AWS Web Application Firewall (WAF): -
AWS WAF: -
AWS Global Accelerator: -

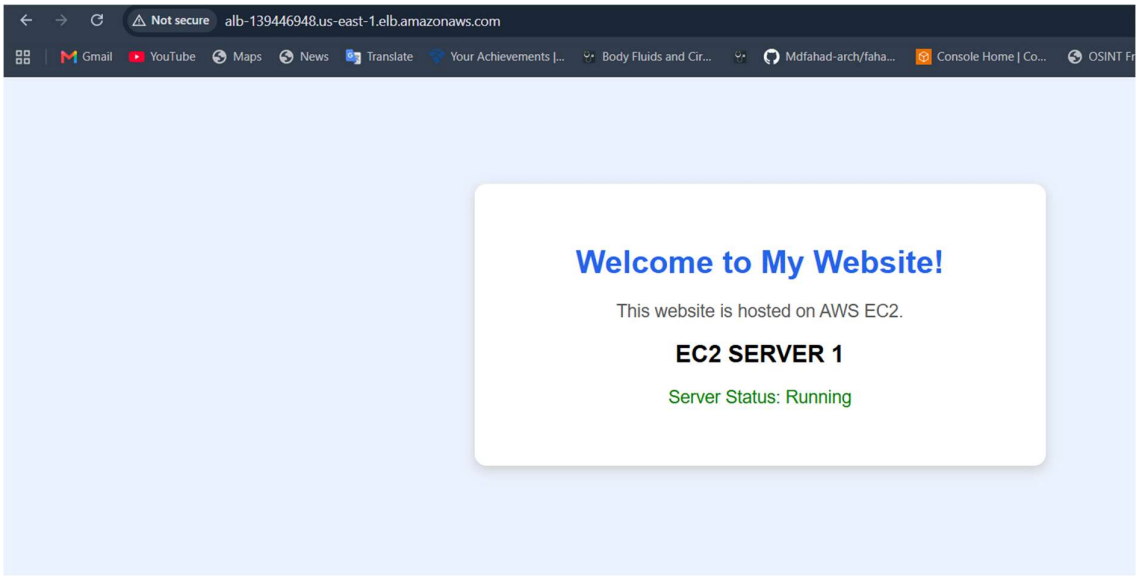
Tags Edit
-

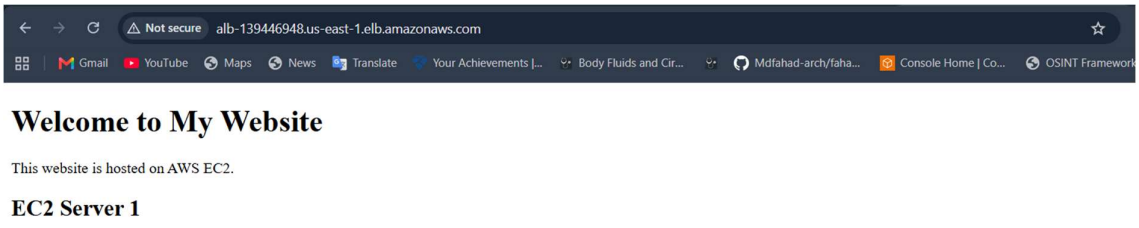
Attributes

ⓘ Certain default attributes will be applied to your load balancer. You can view and edit them after creating the load balancer.









9. Store application load balancer logs in S3.

Step 1:- Create a s3 bucket and create bucket policy.

Step 2:- Now go to loadbalancer, then to attributes scroll to looging.

Step 3:- Click on Access logs and add the s3 bucket location and save it

Step 4:- Now the Load balancer logs are stored in s3 bucket

Create bucket Info

Buckets are containers for data stored in S3.

General configuration

AWS Region

Asia Pacific (Sydney) ap-southeast-2

Bucket type Info

- (•) General purpose
  Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.
- ( ) Directory
  Recommended for low-laten
  of data within a single Availa

Bucket namespace

Choose the namespace where you want to create your bucket. Learn more

- (•) Global namespace
  By default, S3 creates general purpose buckets in the global namespace.
- ( ) Account Regional namespace (recommended)
  General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name Info

alb77-s3bucket

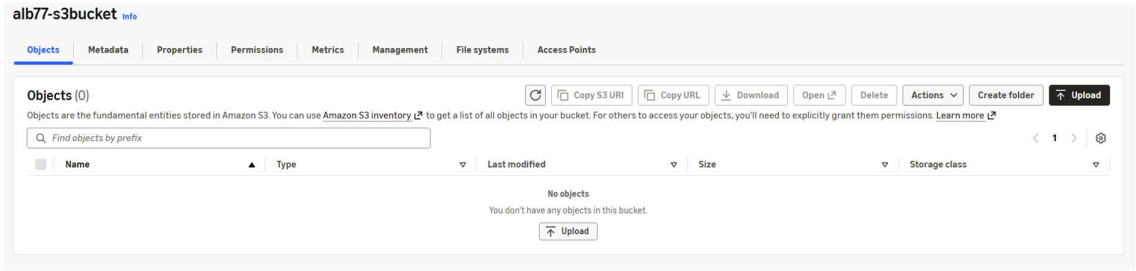
Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.),

Copy settings from existing bucket - *optional*

Only the bucket settings in the following configuration are copied.

Choose bucket

Format: s3://bucket/prefix



Edit bucket policy Info

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. Learn more

Bucket ARN

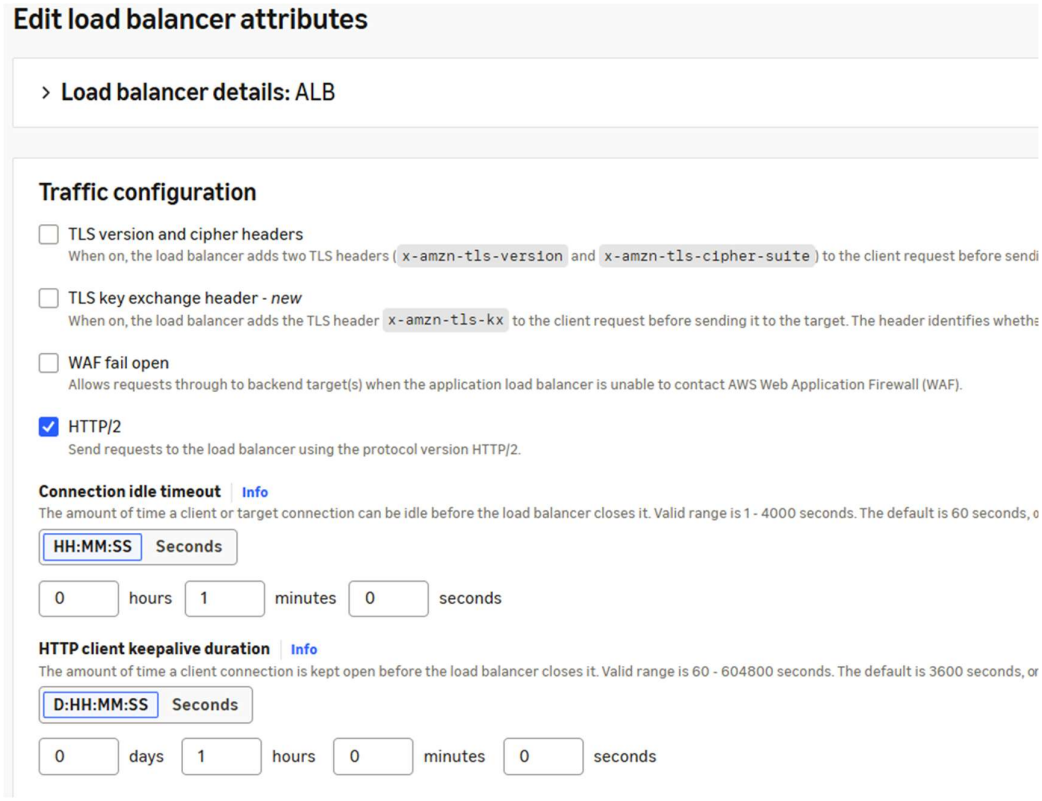
arn:aws:s3:::alb77-s3bucket

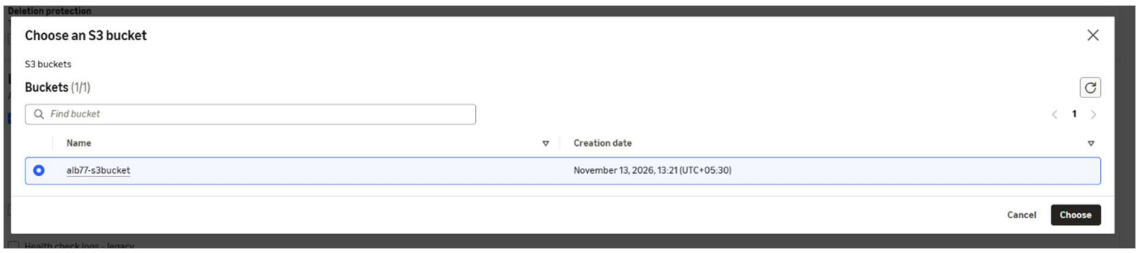
Policy

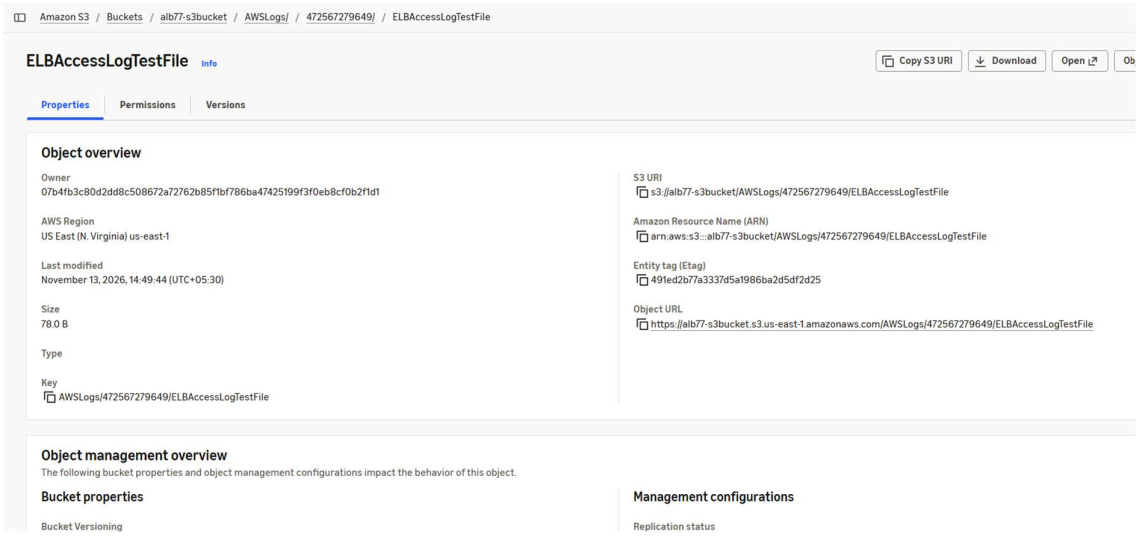
Public access is blocked because Block Public Access settings are turned on for this bucket

To determine which settings are turned on, check your Block Public Access settings for this bucket. Learn more about using Amazon S3 Block Public Access

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "Service": "logdelivery.elasticloadbalancing.amazonaws.com"
      },
      "Action": "s3:PutObject",
      "Resource": "arn:aws:s3:::alb77-s3bucket/AWSLogs/472567279649/*"
    }
  ]
}
```





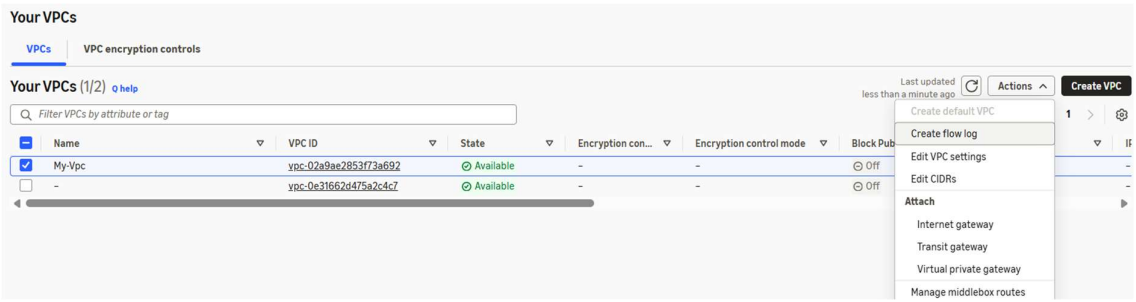


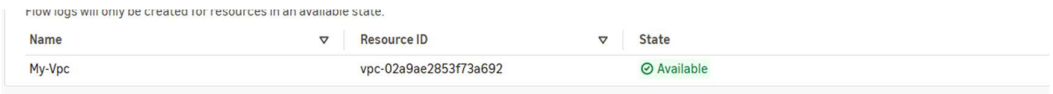
10. Store the VPC flow logs in a CloudWatch log group.

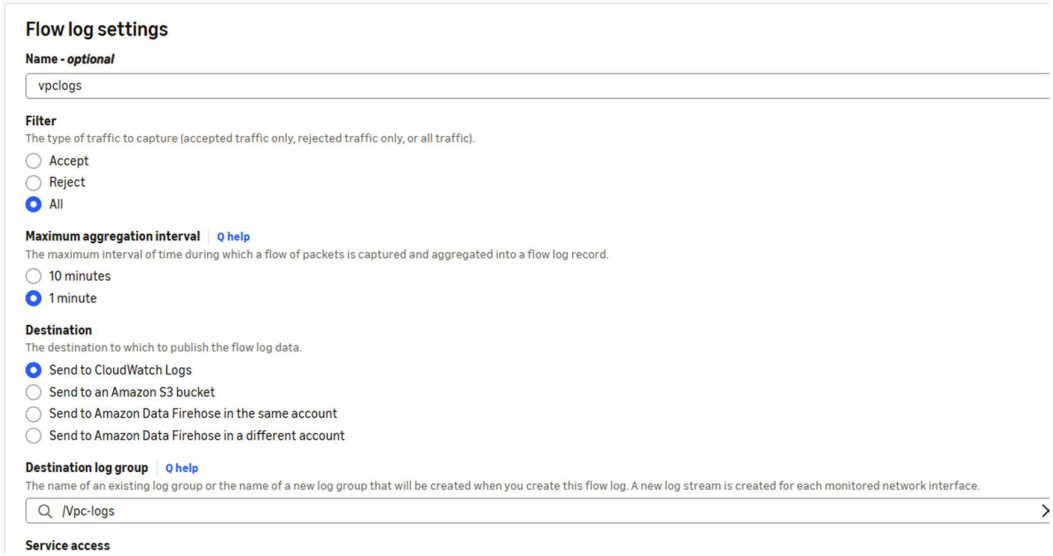
Step 1:- First go to VPC section and select the VPC and click action. There we have Create flow logs.

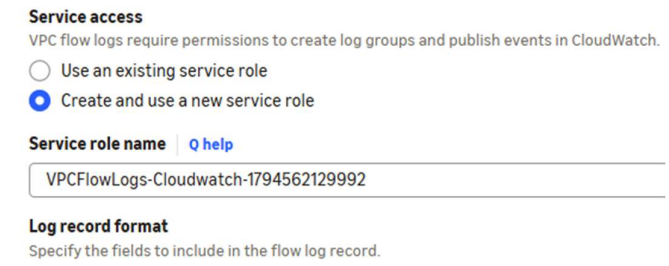
Step 2:- Select the cloudwatch, add the log group name and click on create new role.

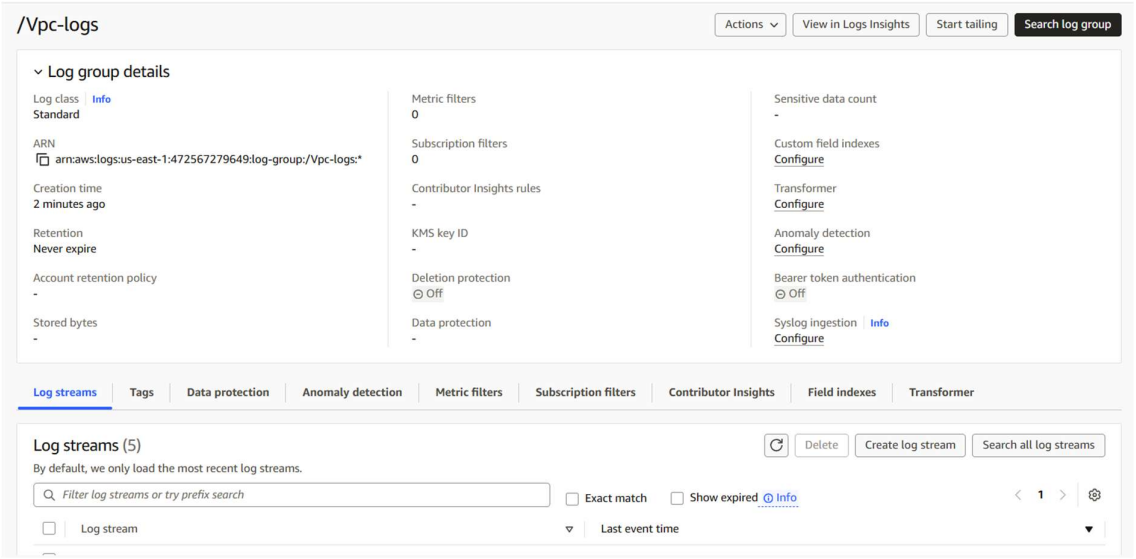
Step 3:- Then click create, we can see the cloudwatch link.











11. Create monitoring dashboards to monitor CPU utilization and to monitor the Apache service.

Step 1:- First go into the instance and create a bash script to monitor the status of the HTTPD.

Step 2:- Change th permissions of the file, start the HTTPD package with systemctl command. Now run the script file.

Step 3:- Create a cronjob to continuously run the monitor script.

Step 4:- Go to cloudwatch service page and create a dashboard. Select the Httpd service metric and CPU utilization of the instance.

Step 5:- Now we can monitor HTTPD service status and CPU utilization from the cloudwatch dashboard.

```
[root@ip-192-60-0-96 ~]# vi Manage-apache.sh
[root@ip-192-60-0-96 ~]# ./Manage-apache.sh
-bash: ./Manage-apache.sh: Permission denied
[root@ip-192-60-0-96 ~]# chmod 777 Manage-apache.sh
```

```
#!/bin/bash

SERVICE="httpd"
NAMESPACE="HTTPD"

# Get EC2 Instance ID (IMDSv2)
TOKEN=$(curl -s -X PUT \
  http://169.254.169.254/latest/api/token \
  -H "X-aws-ec2-metadata-token-ttl-seconds: 21600")

INSTANCE_ID=$(curl -s \
  -H "X-aws-ec2-metadata-token: $TOKEN" \
  http://169.254.169.254/latest/meta-data/instance-id)

REGION=$(curl -s \
  -H "X-aws-ec2-metadata-token: $TOKEN" \
  http://169.254.169.254/latest/meta-data/placement/region)

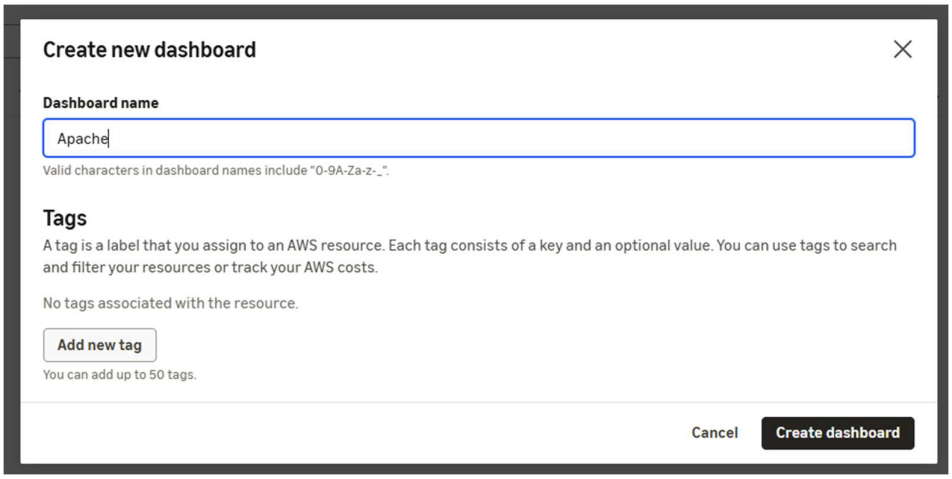
# Check service status
if systemctl is-active --quiet "$SERVICE"; then
    STATUS=1
    echo "Httpd is Running"
else
    STATUS=0
    echo "Httpd is Stopped"
fi

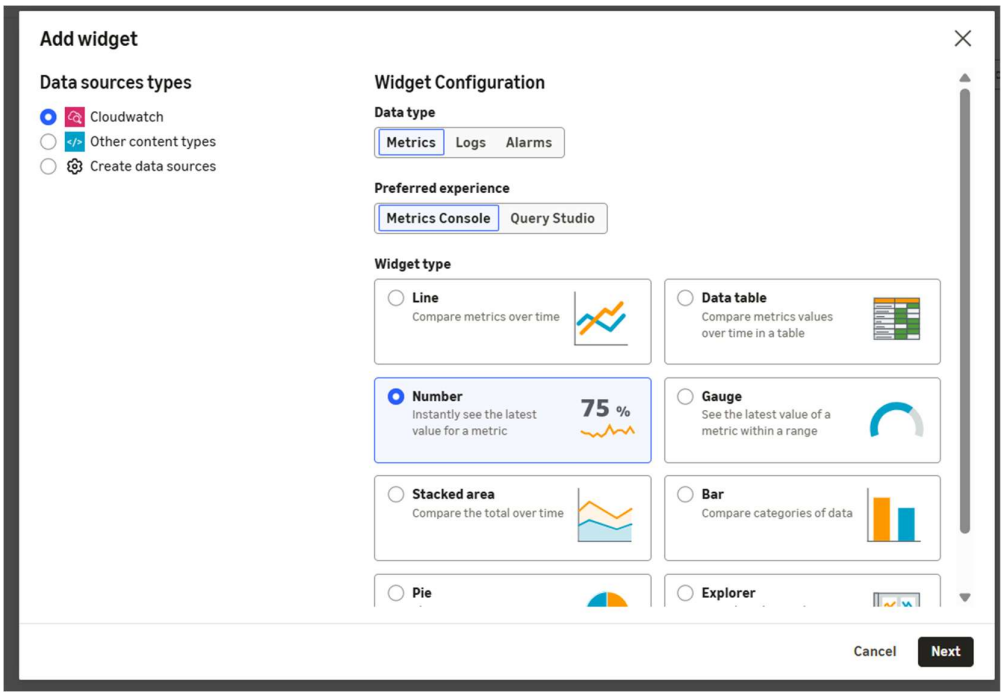
# Send status to CloudWatch
aws cloudwatch put-metric-data \
  --namespace "$NAMESPACE" \
  --metric-name "HttpdServiceStatus" \
  --dimensions InstanceId="$INSTANCE_ID" \
  --value "$STATUS" \
  --unit Count \
  --region "$REGION"

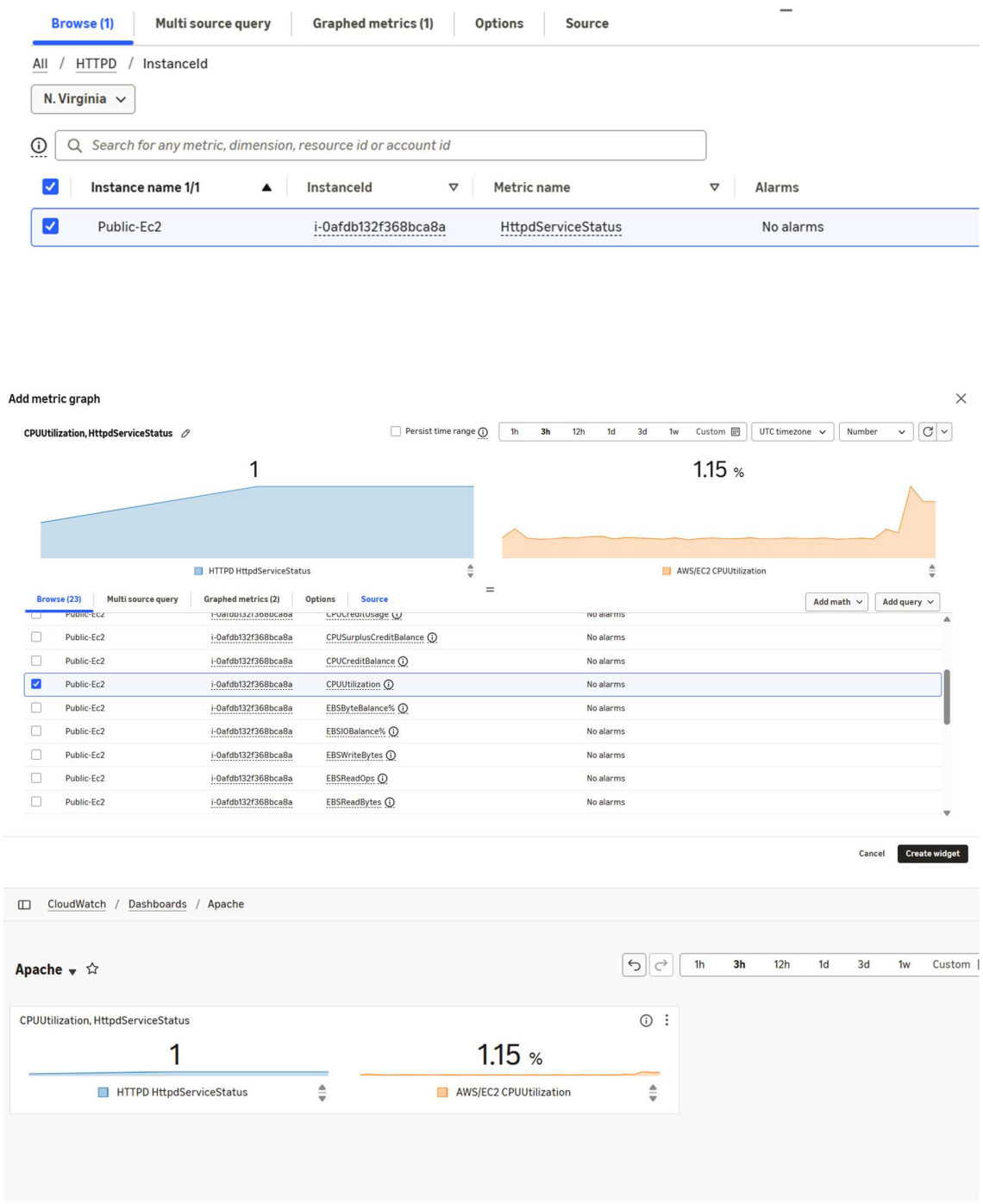
if [ $? -eq 0 ]; then
    echo "Service status sent to CloudWatch"
else
    echo "Failed to send service status"
fi
```

```
[root@ip-192-60-0-96 ~]# systemctl start httpd
[root@ip-192-60-0-96 ~]# ./Manage-apache.sh
Httpd is Running
Service status sent to CloudWatch
```

```
[root@ip-192-60-0-96 ~]# crontab -e
no crontab for root - using an empty one
crontab: installing new crontab
[root@ip-192-60-0-96 ~]# crontab -l
* * * * * ~/Manage-apache.sh
```







12. If CPU utilization is more than 70%, then it should trigger auto scaling and launch new instance.

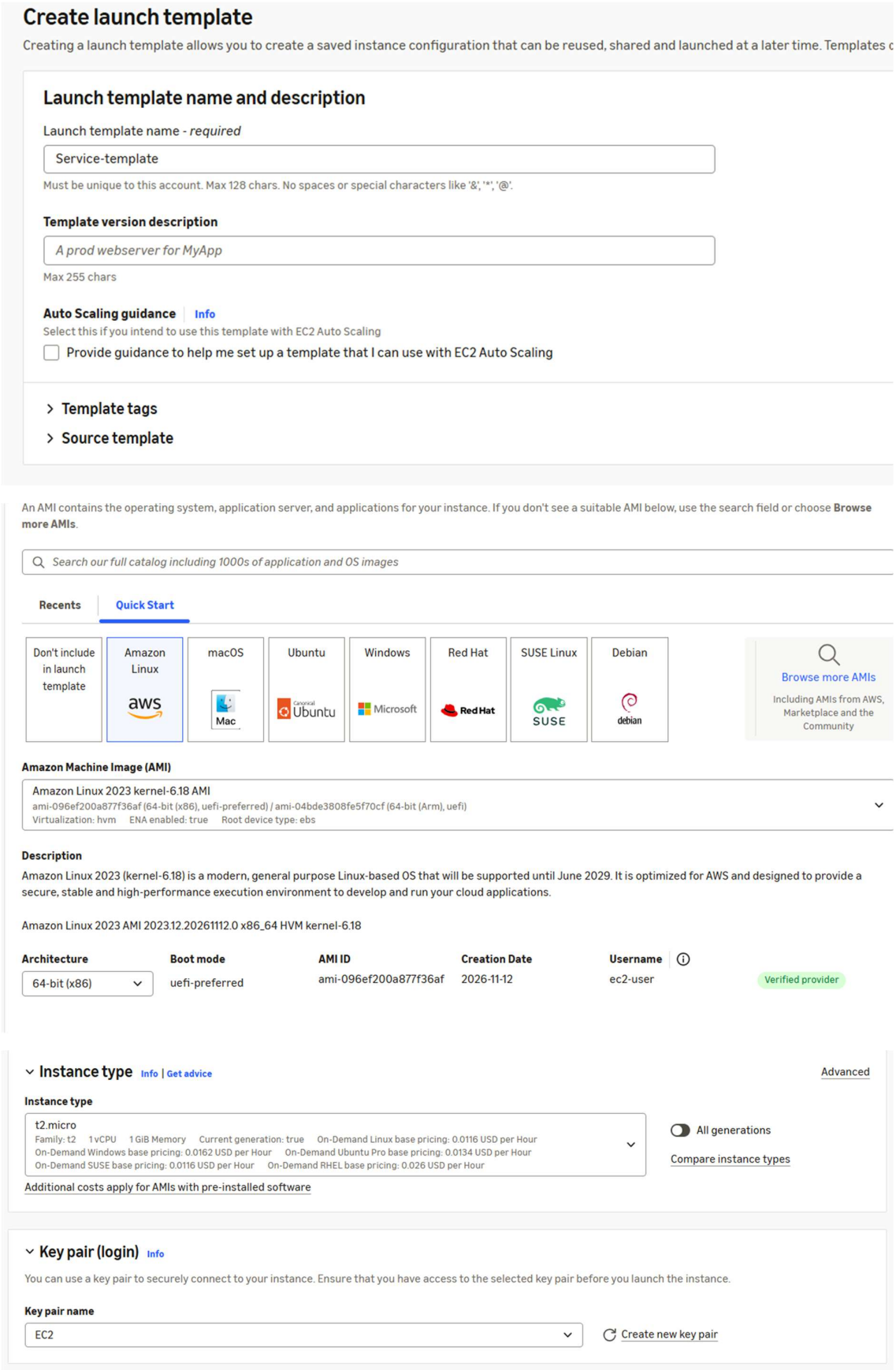
Step 1:- First create a launch template with correct VPC, SG and enable IPv4 Address.

Step 2:- Then go to ASG section and create a default ASG with desired instance 2, min instance 1 and max as 4.

Step 3:- After creating the ASG go to automatic scaling section and create a new dynamic policy where target value should be 70. Then create the Target policy.

Step 4:- Now open the newly created instance and run stress check after some time a new instance is launched.

Step 5:- Go to ASG and then to activity we can see the activities in the ASG.



Network settings Info

Subnet Info

Don't include in launch template

Create new subnet

When you specify a subnet, a network interface is automatically added to your template.

Availability Zone Info

Don't include in launch template

Enable additional zones

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Select existing security group | Create security group

Security groups Info

Select security groups

Compare security group rules

default sg-0386961bbb418583e
VPC: vpc-0e31662d475a2c4c7

Advanced network configuration

Launch template Info

For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

Service-template

Create a launch template

Version

2

Create a launch template version

Description service-2	Launch template Service-template lt-0f448da998a6fcc31	Instance type t2.micro
AMI ID ami-096ef200a877f36af	Security groups -	Request Spot Instances No
Key pair name EC2	Security group IDs sg-052d09390de365dc9	

Additional details

Storage (volumes) -	Date created Fri Nov 13 2026 16:14:03 GMT+0530 (India Standard Time)

Cancel Next

Network Info

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and de

VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-02a9ae2853f73a692 (My-Vpc)
192.60.0.0/16

Create a VPC

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

use1-az2 (us-east-1b) | subnet-08f27fd9915a80d41 (public-01)
192.60.0.0/24

Create a subnet

Availability Zone distribution - *new*

Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

Balanced best effort
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

Balanced only
If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

Reser
Priorit
AZs wi
falls ba

Integrate with other services - *optional* Info

Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing Info

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

Select Load balancing options

- ○ No load balancer
  Traffic to your Auto Scaling group will not be fronted by a load balancer.
- ◉ Attach to an existing load balancer
  Choose from your existing load balancers.
- ○ Attach to a new load balancer
  Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to an existing load balancer

Select the load balancers to attach

- ◉ Choose from your load balancer target groups
  This option allows you to attach Application, Network, or Gateway Load Balancers.
- ○ Choose from Classic Load Balancers

Existing load balancer target groups

Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups ⌄ ↻

Loadbalancer | HTTP ×
Application Load Balancer: ALB

Configure group size and scaling - *optional* Info

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size Info

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances) ⌄

Desired capacity

Specify your group size.

2

Scaling Info

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

1

Equal or less than desired capacity

Max desired capacity

5

Equal or greater than desired capacity

Automatic scaling - *optional*

Choose whether to use a target tracking policy Info

Auto Scaling groups (1) Info

Last updated less than a minute ago ↻ | Launch configurations | Launch templates ↗ | Actions ⌄ | **Create Auto Scaling group**

Search your Auto Scaling groups

‹ 1 › ⚙

Name	Status - *new*	Launch template/configuration ↗	Instances	Instance health - *new*	Desired capacity	Min	Max	Availability Zones
Servers	⊙ Updating capacity	Service-template | Version 2	0	-	2	1	5	use1-az2 (us-east-1

Create dynamic scaling policy

Policy type

Target tracking scaling ⌄

Scaling policy name

Target Tracking Policy

Metric type Info

Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

Average CPU utilization ⌄

Target value

70

Instance warmup Info

60 seconds

☐ Disable scale in to create only a scale-out policy

Activity history (2)

Filter activity history | Filter by a date and time range | 1

Status	Description	Cause	Start time	End time
Successful	Launching a new EC2 instance: i-04462b59e09128058	At 2026-11-13T10:46:38Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2026-11-13T10:46:41Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2026 November 13, 04:16:42 PM +05:30	2026 November 13, 04:16:48 PM +05:30
Successful	Launching a new EC2 instance: i-0884952f817f57ded	At 2026-11-13T10:46:38Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2026-11-13T10:46:41Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2026 November 13, 04:16:42 PM +05:30	2026 November 13, 04:16:48 PM +05:30

```
[ec2-user@ip-192-60-0-50 ~]$ stress-ng --cpu 2 --timeout 300s
stress-ng: info:  [26710] setting to a 300 second (5 mins, 0.00 secs) run per stressor
stress-ng: info:  [26710] dispatching hogs: 2 cpu
^Cstress-ng: info:  [26710] successful run completed in 57.82s
[ec2-user@ip-192-60-0-50 ~]$
Broadcast message from root@localhost (Fri 2026-11-13 11:05:53 UTC):

The system will power off now!
```

Filter activity history | Filter by a date and time range | 1

Status	Description	Cause	Start time	End time
Successful	Launching a new EC2 instance: i-0a7df7717e9a2b42b	At 2026-11-13T10:54:44Z an instance was launched in response to an unhealthy instance needing to be replaced.	2026 November 13, 04:24:45 PM +05:30	2026 November 13, 04:24:51 PM +05:30
Successful	Terminating EC2 instance: i-0884952f817f57ded	At 2026-11-13T10:54:43Z an instance was taken out of service in response to an EC2 health check indicating it has been terminated or stopped.	2026 November 13, 04:24:43 PM +05:30	2026 November 13, 04:29:46 PM +05:30
Successful	Launching a new EC2 instance: i-0f2205f3da5406ea1	At 2026-11-13T10:52:48Z an instance was launched in response to an unhealthy instance needing to be replaced.	2026 November 13, 04:22:49 PM +05:30	2026 November 13, 04:22:55 PM +05:30
Successful	Terminating EC2 instance: i-04462b59e09128058	At 2026-11-13T10:52:47Z an instance was taken out of service in response to an EC2 health check indicating it has been terminated or stopped.	2026 November 13, 04:22:47 PM +05:30	2026 November 13, 04:27:49 PM +05:30

Instances (6) Info

Last updated 1 minute ago | Connect | Instance state | Actions | Launch instances

Saved filter sets: Choose filter set | Filter Instance by attribute or tag (case-sensitive) | 1

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ad...	Elastic
Private-Ec2	i-099e9d664dff3167c	Running	t3.micro	3/3 checks passed..	us-east-1b	-	-	-
Public-Ec2	i-0afdb132f368bca8a	Running	t3.micro	3/3 checks passed..	us-east-1b	-	18.206.254.2	-
	i-0a7df7717e9a2b42b	Running	t2.micro	2/2 checks passed..	us-east-1b	-	3.82.104.137	-
	i-04462b59e09128058	Terminated	t2.micro	-	us-east-1b	-	-	-
	i-0884952f817f57ded	Terminated	t2.micro	-	us-east-1b	-	-	-
	i-0f2205f3da5406ea1	Running	t2.micro	2/2 checks passed..	us-east-1b	-	18.212.49.254	-